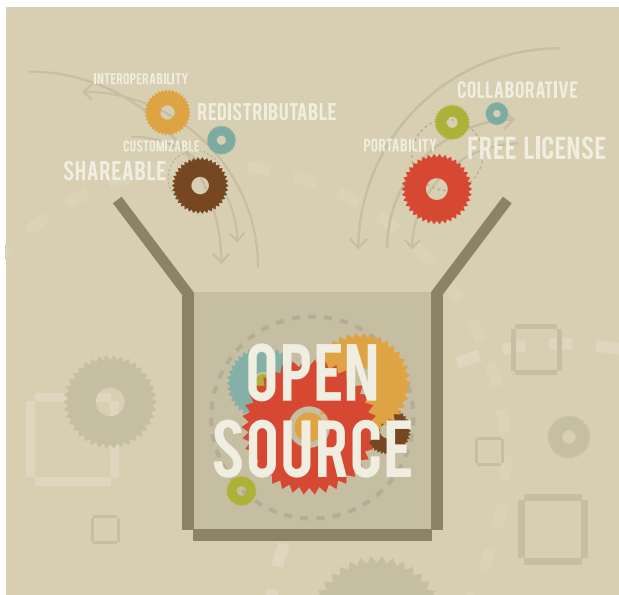


The Benefits and Drawbacks of OSS

Open source software has witnessed a steady increase in adoption over the past two decades, but the emergence of artificial intelligence, blockchain and edge computing has taken its popularity to an all-time high. While OSS has many benefits, it does come with a set of challenges.



A silent revolution is taking place in the technology and software development world, one that encourages collaboration, celebrates transparency, and empowers individuals and organisations. Open source software (OSS) is at the centre of this transformation, reshaping our interactions with technology and the digital universe.

The origins of OSS

OSS has its origins in the late 1970s and early 1980s, when the ethos of ‘free software’ began to take shape. Pioneers such as Richard Stallman, the founder of the Free Software Foundation, laid the groundwork for the OSS movement. Stallman’s GNU Project, which began in 1983, aimed to create a free UNIX-like operating system that included all necessary software. This project was motivated by the belief that software should be freely shared, with users free to run, study, modify, and distribute it as they saw fit.

The term ‘open source’, as we know it today, became popular with the establishment of the Open Source Initiative

(OSI) in 1998. The OSI sought to promote open source software adoption by emphasising the practical benefits of transparency and collaboration. It proposed the open source definition — a set of requirements for software to be considered open source. This definition included principles such as free redistribution, access to source code, and no discrimination against individuals or groups.

The first impact of open source was the liberation of software development from proprietary constraints. This novel approach to software development enabled individuals, organisations, and communities to collaborate on projects on an unprecedented scale. OSS projects sprouted up all over the digital landscape, laying the groundwork for a new era of technology.

The GNU/Linux operating system was one of the first and most influential OSS projects, a collaborative effort to create a free and open source UNIX-like operating system. Linus Torvalds launched Linux in the early 1990s, and it quickly gained traction as a dependable and versatile alternative to proprietary operating systems.

The success of Linux catalysed the adoption of open source practices in a variety of fields, inspiring a wide range of software projects to take an open approach. Apache, an open source web server, became the World Wide Web’s foundation, serving many websites worldwide. The Mozilla project, responsible for the Firefox web browser, marked yet another milestone in spreading open source ideals. These developments demonstrated that not only was open source software feasible, but was also capable of outperforming its proprietary counterparts.

Why open source is important

- **Collaboration and innovation:** OSS is a hotbed of innovation. It promotes global developer collaboration, resulting in faster software development and enhancement. This collaborative approach enables businesses to tap into a vast pool of expertise, which frequently results in cutting-edge features and functionalities.
- **Cost-efficiency:** One of the primary benefits of OSS is the ability to save money. The use of open source software can be a game changer for businesses, particularly startups